NOTES FROM THE RESEARCH (REVIEW ON THE PAPERS)

1. ***A comprehensive update of genotype–phenotype correlations in PMM2-CDG: insights from molecular and structural analyses***

Reference: Oliveira_2025_R1.pdf

Question: How 41 variant missenses can explain the structure and function of the PMM2 and how this is related with the phenotype of the patient?

How the paper resolves it: They take pathogenic variants from ClinVar, analyze them using the 7O4G structure (wild-type PMM2) and the PyMOL mutagenesis tool, and use Briso-Montiano as a basis. They also combine this with data on phenotype, severity, and enzymatic activity.

Data to use:

- PMM2 converts mannose-6-phosphate (M6P) into mannose-1-phosphate (M1P) and requires G16BP or M16BP as an activator.
- M1P is precursor of GDP-mannose, which gives mannose to Dol-P-M (precursor of N-glycosylation)
- PMM2-CDG is an autosomal recessive disease and the most prevalent N-glycosylation disorder). The lack of activity causes hypoglycosylation of many glycoproteins and multiorgan involvement
- The primary impact is on the nervous system (causing ataxia, epilepsy, peripheral neuropathy…)
- More than 1,000 patients described; incidence between 0.06 and 5 per 100,000; approximately 20% die within the first four years; only symptomatic treatment is available.
- 142 variants described, 133 missense; R141H is the most frequent and has never been observed in a homozygous state.
- Most patients are compound heterozygotes (one hypomorphic allele and one loss-of-function allele).
- Six mechanisms of action: activator binding, catalysis, dimerization, cap folding, core folding, and linker 2.

T237M relevant:

- T237M is c.710C>T and is classified as a folding.
- The variant is located in the helix α9, in the core domain.
- They propose that the change in amino acids could break the hydrogen bonds, which could cause and affect the catalysis and the folding stability.
- The experimental data cited: Yuste-Checa et al. indicate that T237M has a high dimerization capacity and intermediate catalytic activity.
- They also cite compound cases (R141H/T237M) (Imtiaz) and T237M with a deletion-insertion (Tayebi).

Limitations:

- This is a review so its structural conclusions are primarily predictions based on a static structure, rather than new experiments.
- The authors note that dimerization and the protein's half-life complicate the correlations (pp. 13–14).
- Much of their classification is derived from Briso-Montiano (OP4), so it cannot be treated as an independent source.
- They only include variants classified as pathogenic in ClinVar.

2. ***Insight on molecular pathogenesis and pharmaco chaperoning potential in phosphomannomutase 2 deficiency, provided by novel human phosphomannomutase 2 structures***

Reference: Briso-Montiano_2021_OP4.pdf

Question: What is the complete structure of the human PMM2, what is said about the effect of patient variants and which ones could be treated with pharmaco chaperoning?

How the paper resolves it: They produce recombinant PMM2 WT and T237M. They crystallize them both with and without glucose-1,6-bisphosphate (Glc-1,6-P_2) and determine their structures via X-ray crystallography. They also include SEC-MALS assays (to confirm the dimer) and isothermal titration calorimetry (ITC) (pp. 3–5).

Data to use:

- PMM2 is essencial. Its lack of activity is lethal for knockout rats and there are no homozygotes for null alleles such as R141H.
- PMM1 does not compensate for the lack of PMM2, even though it catalyzes the same reaction.
- More than 135 pathogenic variants; R141H and F119L are the most frequent.
- 66% sequence identity between PMM1 and PMM2; PMM1 exhibits phosphatase activity toward the activator, whereas PMM2 does.
- A dimer is a protein composed of two chains (subunits) joined together. Each chain is a complete protein molecule, and together they form the functional unit. There are two types: Homodimers and Heterodimers. PMM2 is only active as a dimer, specifically the homodimer type, which means the two chains are identical. The two chains join at the "cap" domain, which serves as the dimerization interface.
- 246-aa proteins that form stable dimers (WT and T237M); crystals at ~2 Å (T237M) and 2.7–3.1 Å (WT)
- The chain divides in zones depending on the amino acid position. (number 1 is the first, 246 the last). Said zones are:
    - Core domain: amino acids 1 to 83 and 189 to 246. It is a single folded unit, but with a piece at each end.
    - Cap domain: amino acids 87 to 184, in the middle.

    - Link 1 and 2: Amino acids 84 to 86 and 185 to 188. Two short and flexible sections connecting the cap to the core.
- Different conformation: In the dimer, the two chains are not identical in 3D. One of the chains has the core closer to the cap than in the other. This could be expected from an enzyme that opens and closes to trap its substrate, a process facilitated by the linkers.
- Mg-1(catalytic) and Mg-2(structural) are two magnesium ions inside the core.
- Mg-1 participates in the reaction, it is held in place by two loops: the one spanning amino acids 12–24 and the one spanning 212–217. This is where T237 comes into play, as its side chain bonds with T16, which is located in the first of these loops.
- Mg-2 does not participate in the reaction, only helps to maintain the form.
- The dimer interface is the interface through which the two chains are joined. It occupies 910 Å² per chain, and at its center is a chloride ion that helps hold them together. This is relevant for dimerization variants, but not for T237M.
- About the experiment, the authors took all the patient variants (101) to study in which position were located. They cover 79 of the 246 amino acids and 93 were classified as mechanisms. The majority of them are located in important zones such as the loops that hold the Mg-1.
- Pharmaco Chaperones are small molecules that help a misfolded protein fold better or degrade more slowly. The authors suggest that approximately 80% of the variants in the core and 50% of those in the cap could be candidates. Since T237M affects the folding of the core, it might be among them.

T237M relevant:

- They choose to study T237M because it is a frequent variant in Spanish patients with an altered folding and stability. (p3). Is also cited as an example of a variant with more than one impact mechanism. (p13)
- Is located on the α9 helix, at the C-terminal end of the core (p. 7, Fig. 2).
- The side chain of T237 forms a hydrogen bond with T16, which is located in the β1-α1 loop that coordinates Mg-1. Losing this interaction would explain the reduced stability and activity. (Cause the methionine does not have -OH)
- When the complete structures of WT and T237M (with and without Glc-1,6-P_2) are superimposed, the overall folding remains virtually unchanged. The dimers are essentially identical, with low RMSD values (a measure of how much one structure deviates from another).
- Gravity: They describe it as a “relatively mild” change and discuss a G176V/T237M patient whose survival could be due to the partial activity of T237M (p. 12).
- Structures: 7O5Z (T237M without ligand, 2.07 Å) and 7O58 (T237M with Glc-1,6-P_2, 1.97 Å). Those of WT are 7O0C, 7O1B and 7O4G (p. 4, Table 1).

Limitations:

- This is a structural study: crystal structures are static and, in the case of T237M, show no visible changes. The effect is inferred based on the loss of a bridge, as well as previous studies on stability and activity.

- The classification of 93 variants is predicted based on the structure. The authors themselves note that a single variant may act through multiple mechanisms.
- The variants are sourced from the HGMD; the authors point out that hypomorphic alleles may be overrepresented, as a certain level of activity is required for survival.
- The activity and stability of T237M are attributed to previous research; they are not measured in this study (with the exception of confirming that it forms stable dimers).